AMRITA VISHWA VIDYAPEETHAM
SCHOOL OF COMPUTING
DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

Motion Consistency Evaluation in Unmodified Surveillance Video

23CSE399 – Project Phase 1 (Panel Review 1)

Team Members		Guide	Project Type	Marks
CB.SC.U4CSE23339	Nanditha C	**Dr. N Lalithamani** *Associate Professor,* *Dept. of CSE*	**Type 1** Research	.../ 5
CB.SC.U4CSE23353	Krishna Veni Valluri			
CB.SC.U4CSE23358	Vinutha Yartha			
CB.SC.U4CSE23361	Harisree M			

1 Project Proposal and Objectives

Project Proposal

The proposed project is based on the evaluation of motion consistency in surveillance video feeds to ensure the reliability of video cameras. In practice, video feeds from cameras can be affected by problems like freezing or inconsistent motion, which are not easily detectable due to the assumption of the feed being consistently available.

The proposed approach for detecting anomalies in video feeds involves the evaluation of temporal motion patterns using a dual-stream LSTM autoencoder, which combines frame differencing and optical flow.

Objectives

- Detect frozen or irregular video feeds
- Remove dependency on threshold values
- Classify video feeds as healthy, frozen, or irregular
- Provide a continuous reliability score
- Ensure the approach can handle multiple cameras

2 Introduction and Background

Surveillance systems are commonly used in scenarios like traffic monitoring and public safety, wherein a continuous video feed is required from a number of cameras.

The biggest challenge in surveillance systems is ensuring that the feed from all cameras is working properly. In a real-world scenario, a frozen video feed or unusual motion can occur without any alert being sent out.

The need for this research is to identify hidden failures in video feeds, particularly in scenarios wherein manual monitoring is not possible.

The existing research mainly deals with activity recognition, anomaly detection, and video tampering detection using motion-based features and deep learning techniques. However, all these approaches assume that the video feed is working properly.

The need for this project is to ensure that the video feed is working properly by using a camera health monitoring approach based on motion consistency analysis.

3 Execution Plan

Required Subdomains

The project requires that we know about various domains. These are:

- Computer Vision – Motion Analysis
- Temporal Modeling – LSTM

- Optical Flow Methods
- Image Preprocessing
- Anomaly Detection
- Similarity Measurement (SSIM)

Mapping Subdomains to Components

The various subdomains that we are going to work with are given below:

Subdomain	Component	Role
Motion Analysis	Frame Differencing	Captures the changes in the pixels
Optical Flow	Farneback Method	Captures the amount of motion in the frames
Temporal Modeling	LSTM Autoencoder	Analyzes the patterns of motion
Preprocessing	Resize + Normalize	Standardizes the input images
Similarity Check	SSIM	Detects repetition in the frames
Guard Mechanism	Pixel Variance	Detects low-motion sequences
Anomaly Detection	Divergence Anomaly Score	Identifies inconsistencies between streams
Decision System	Adaptive Threshold	Classifies feed

Learning Plan

The learning plan is also divided into various stages. These are:

- Firstly, we learn about image processing, motion analysis, and LSTM networks.
- Then we learn about optical flow and anomaly detection.
- Next, we learn about advanced concepts such as dual-stream modeling, divergence-based scoring, and adaptive thresholding to integrate the overall complete system.

4 Proposed Methodology and Architecture

The system that is being proposed makes use of a learning-based approach that is able to assess motion consistency in videos obtained from a surveillance feed.

The videos that are input are preprocessed to resize and normalize the videos and convert them to grayscale. Guard layers are implemented that utilize variance and SSIM to identify frozen videos.

For videos that are not frozen, motion features are extracted using differencing and optical flow. These are used to extract two motion streams. These are then input into a dual-stream LSTM autoencoder that is able to identify normal motion patterns.

The reconstruction errors are combined using a divergence anomaly score. A dual-window threshold is used to identify if a video is healthy, frozen, or irregular. A feed reliability score is also obtained.

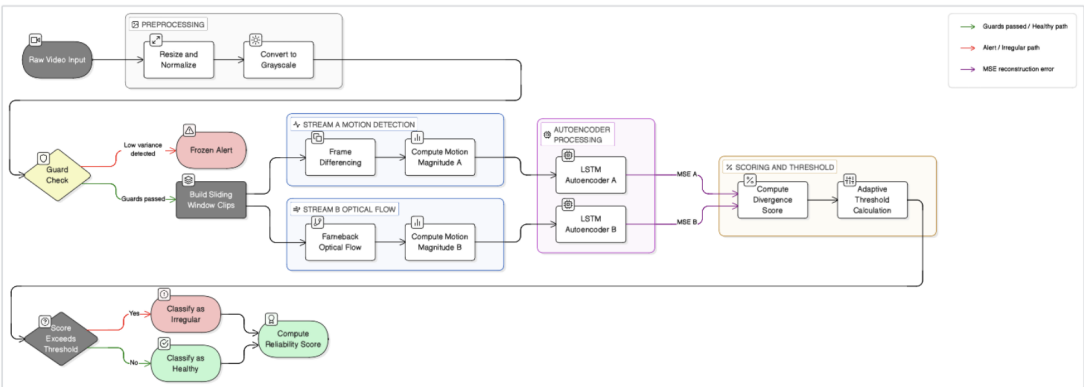

Figure 1: Architecture Diagram

Architecture Explanation

- **Preprocessing:** This layer normalizes the input feeds and minimizes complexity
- **Guard Layer:** This layer detects frozen feeds using pixel variance and SSIM
- **Feature Extraction:** This layer creates motion feeds using differencing and optical flow
- **Temporal Modeling:** This layer learns motion patterns using dual-stream LSTM autoencoder
- **Anomaly Scoring:** This layer uses the divergence score for detecting anomalies
- **Decision Module:** This layer uses adaptive thresholding for feed classification and reliability

Design Justification

The proposed system does not require object detection, making it a lightweight solution. It is also scalable for multiple cameras using the module-based structure.

5 Fundamentals Learned

- Developed an understanding of image preprocessing operations such as resizing, normalization, and grayscale transformations
- Developed an understanding of basic motion analysis through differencing and optical flow
- Developed an understanding of the fundamentals of LSTM in the context of handling time-series data
- Developed an understanding of the autoencoder concept in the context of anomaly detection
- Developed an understanding of similarity measurement through SSIM
- Developed a basic understanding of classification through thresholding

6 Work Completed, Research Findings and Future Plan

Work Completed

- Initially, a methodology based on a statistical motion consistency framework with a sliding window approach and threshold detection was proposed. Further, a more refined approach was developed, moving beyond a statistical approach.
- A literature survey was done on video anomaly detection, motion, and video quality in surveillance.
- The gaps in the research are identified as the lack of reliable camera feed sources and the use of threshold detection.
- The essential research concepts have been studied, including frame differencing, optical flow, temporal consistency, and anomaly detection.
- The system approach to motion consistency evaluation and camera feed reliability scoring was developed.
- The system architecture for surveillance feed health was designed.

Research Findings

- Existing systems are not focused on the camera health monitoring
- Existing systems are based on manual thresholding, and it is not scalable
- Learning-based systems provide more scalability and flexibility
- Dual stream validation is more effective than single stream validation

Future Plan

Next Phase:

- Pre-processing and Guard layer implementation
- Dual LSTM autoencoder training
- Divergence scoring mechanism development

Final Phase:

- Evaluation of the system with artificial fault
- Performance evaluation of the system using standard parameters
- System integration and final report

References

[1] G. Pang, H. Wang, J. Cheng, C. Zhang, H. Wang, and D. Doermann, "Video anomaly detection in 10 years: A survey and outlook," *IEEE Transactions on Pattern Analysis and Machine Intelligence (TPAMI)*, 2024, arXiv preprint arXiv:2404.16480.

[2] X. dong Bi, X. He, S. Xiong, Z. Zhao, H. Chen, and R. E. Sheriff, "Blind video quality assessment based on spatio-temporal feature resolver," *Neurocomputing*, vol. 575, p. 127264, 2024.

[3] H. Liu *et al.*, "Advancing video anomaly detection: A concise review and a new dataset," in *Advances in Neural Information Processing Systems (NeurIPS)*, 2024.

[4] L. Li, J. Lu, S. Zhang, L. Mohaisen, and M. Emam, "Frame duplication forgery detection in surveillance video sequences using textural features," *Electronics*, vol. 12, no. 22, p. 4597, 2023.

[5] S. B. Veesam, B. T. Rao, Z. Begum, R. S. M. L. Patibandla, A. A. Dcosta, S. Bansal, K. Prakash, M. R. I. Faruque, and K. S. Al-Mugren, "Multi-camera spatiotemporal deep learning framework for real-time abnormal behavior detection in dense urban environments," *Scientific Reports*, vol. 15, no. 1, p. 26813, 2025.

[6] G. Mittag, B. Naderi, V. Gopal, and R. Cutler, "Lstm-based video quality prediction accounting for temporal distortions in videoconferencing calls," *arXiv preprint arXiv:2303.12761*, 2023.

[7] N. Akhtar, M. Hussain, and Z. Habib, "Two–stage detection and localization of inter–frame tampering in surveillance videos using texture and optical flow," *Mathematics*, vol. 12, no. 22, p. 3482, 2024.

[8] X. Zhang *et al.*, "Anomaly detection method of surveillance video based on global-local information," *ResearchGate*, 2023.

[9] M. P. Priya and K. UlagaPriya, "Video prediction based on temporal aggregation and recurrent propagation for surveillance videos," *ResearchGate*, 2025.

[10] S. Hussain *et al.*, "Engineering a multi-sensor surveillance system with secure alerting for next-generation threat detection and response," *Scientific Reports / NIH*, 2024.

[11] B. Liu *et al.*, "Surveillancevqa-589k: A benchmark for comprehensive surveillance video-language understanding with large models," *arXiv preprint arXiv:2505.12589*, 2025.

[12] S. Ali *et al.*, "Robust motion detection and classification in real-life scenarios using motion vectors," *PLOS ONE*, 2023.

[13] Y. Zhang *et al.*, "Video quality assessment with texture information fusion for streaming applications," *arXiv preprint arXiv:2407.01234*, 2024.

[14] J. Wang *et al.*, "Frame shuffling forgery detection method for mpeg-coded video," *IEEE Access*, 2023.

[15] S. Sarker *et al.*, "Proactive anomaly surveillance system for cctv footage analysis in adverse environmental conditions," *Scientific Reports*, 2024.

[16] J. Vora *et al.*, "Securecam: Selective detection and encryption enabled application for dynamic camera surveillance videos," *IEEE Internet of Things Journal*, 2023.